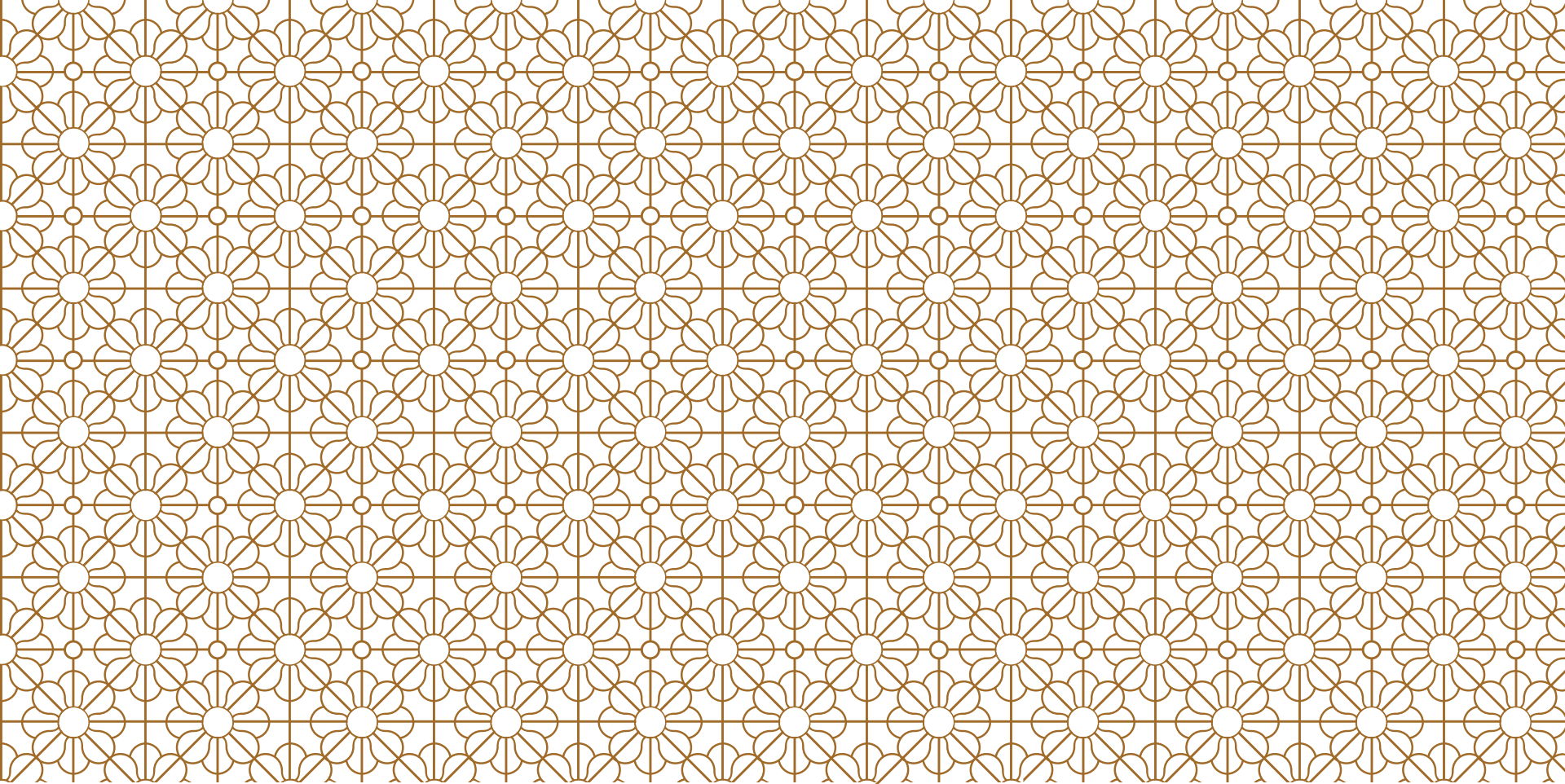




FUNCTION

INTRODUCTION TO FUNCTIONS

Nyimas Dewi Sartika
Universitas Multimedia Nusantara
2021 - 2022

Introduction to functions

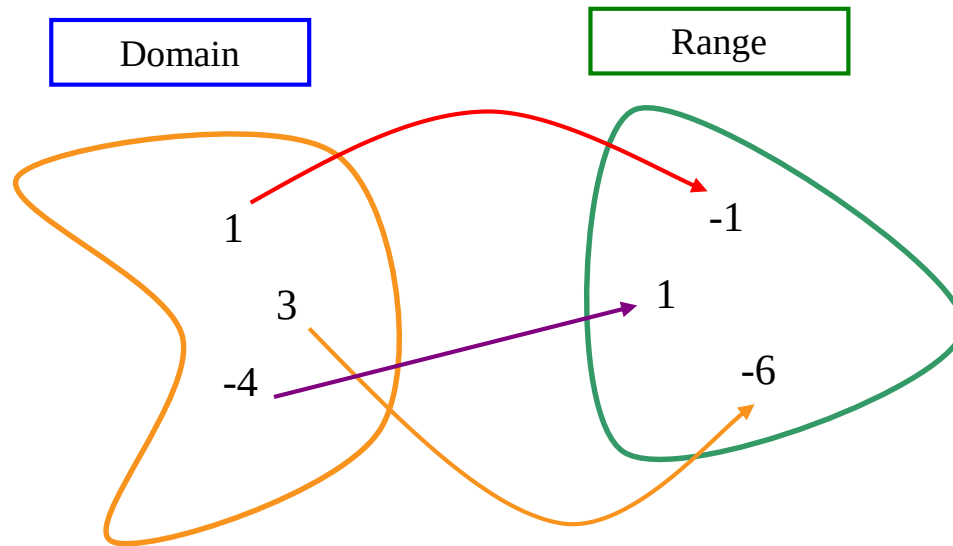
Four examples of functions:

- ▶ The area of a circle depends on the radius: $A = \pi r^2$.
- ▶ Population of the world depends on time: $P(1950) = 2,560,000,000$.
- ▶ The cost of posting a package depends on the weight: $C(w)$.
- ▶ The vertical ground acceleration during an earthquake: $S(t)$.

Domain and range of functions

A *function* f is a rule which assigns to each element x in a set D exactly one element, $f(x)$, in a set E .

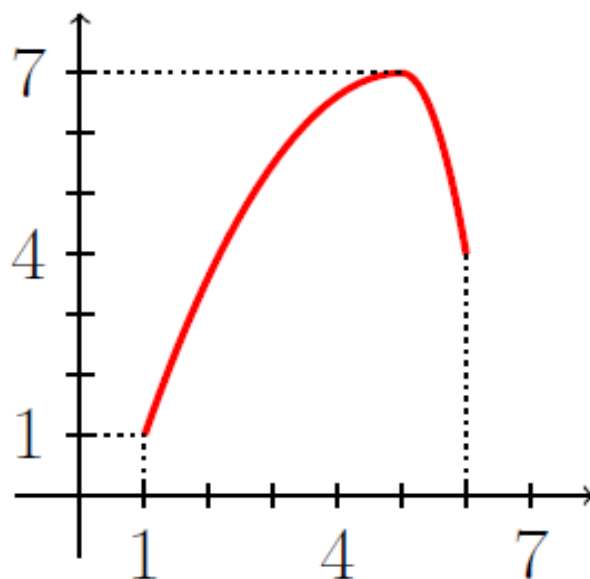
The set D is the *domain* of f . The *range* of f is the set of all possible values of $f(x)$ as x varies through the domain.



A symbol representing an arbitrary element of the domain is called an *independent variable* and a symbol representing an arbitrary element of the range is a *dependent variable*. In the example of the circle: r is the *independent variable* while A is the *dependent variable*.

Graphs of functions: a common way of representing a function is by a graph.

Formally, the graph of the function f is the set of ordered pairs $\{ (x, f(x)) \mid x \in D \}$.



$$\text{Dom}(f) = [1, 6] \quad \text{and} \quad \text{Ran}(f) = [1, 7]$$

Function Notation

$y = 3x^2 + 2$ is a *function*, with values of x as the **domain** and values of y as the **range**.

We write $f(x)$ in place of y : $f(x) = 3x^2 + 2$

Find $f(-2)$, $f(0)$, and $f(b+1)$, where $f(x) = 3x^2 + 2$.

$$f(-2) = 3(-2)^2 + 2 = 3(4) + 2 = 14$$

$$f(0) = 3(0)^2 + 2 = 2$$

$$f(b+1) = 3(b+1)^2 + 2 = 3(b^2 + 2b + 1) + 2 = 3b^2 + 6b + 5$$

Representations of functions

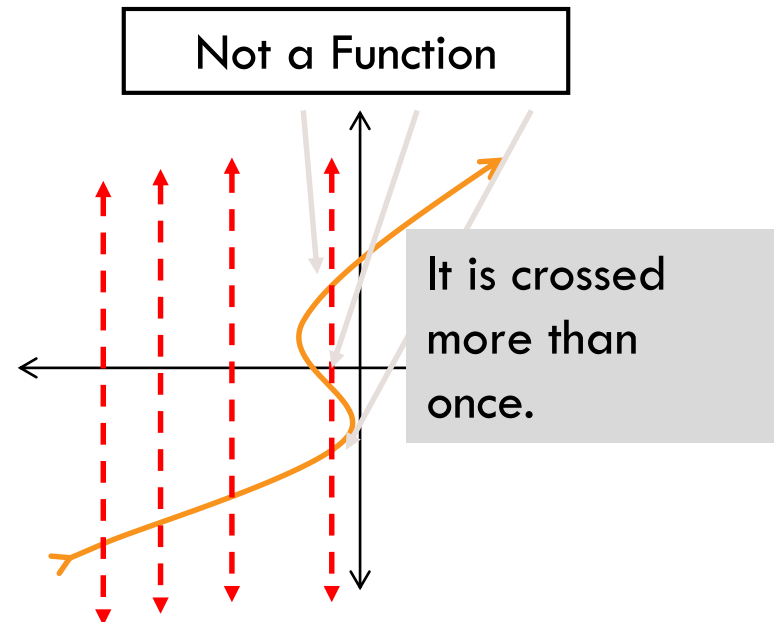
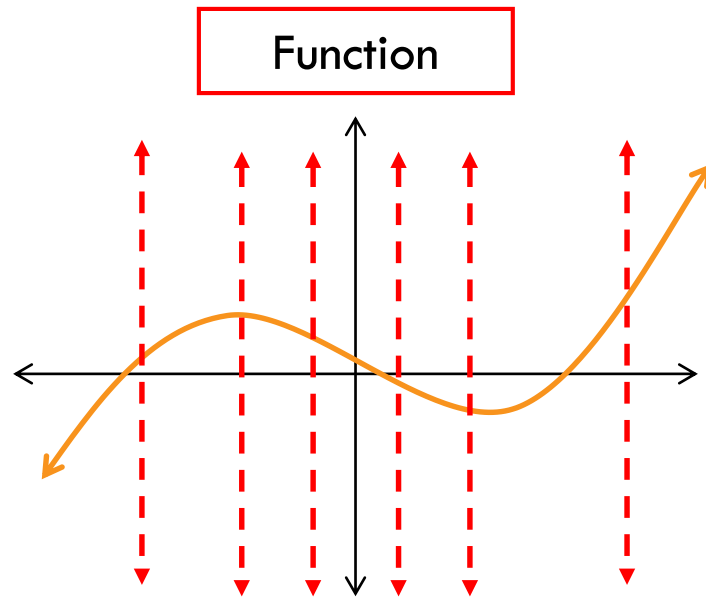
There are four ways to represent a function:

- ▶ Verbally (describe in words)
- ▶ Tables
- ▶ Graphically
- ▶ Algebraically

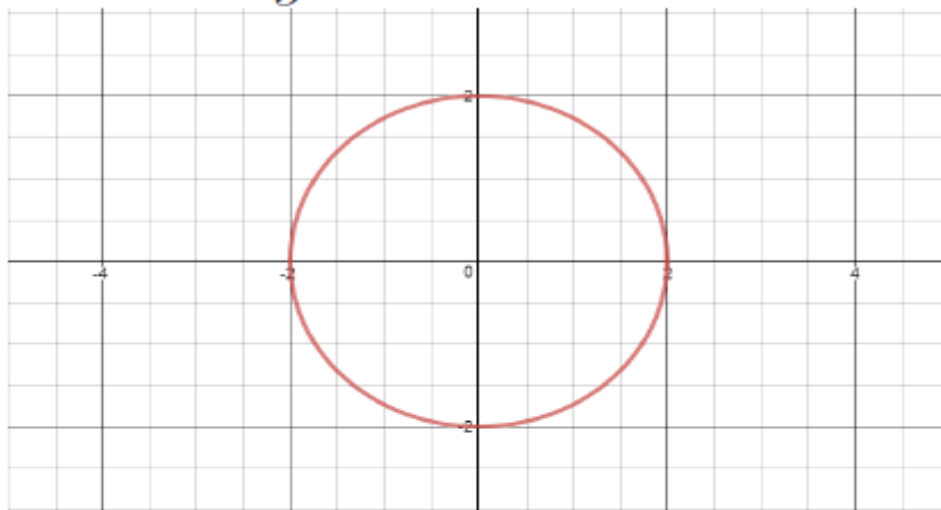
The vertical line test

How do we know if a curve is a function?

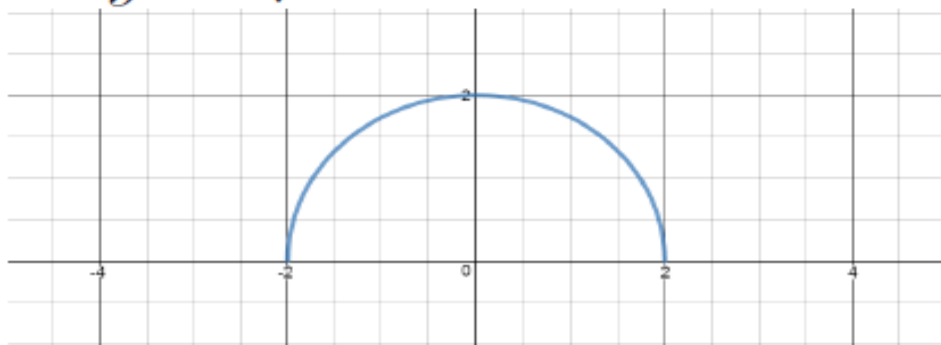
A curve in the xy -plane is a function of x if and only if *no vertical line intersects the curve more than once*.



The curve $x^2 + y^2 = 4$ is *not* a function:



The curve $y = \sqrt{4 - x^2}$ is a function:



Piecewise defined functions

We are already familiar with one example of a piecewise defined function, the absolute value function:

$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Now sketch the function

$$f(x) = \begin{cases} -x + 2 & \text{if } x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$$

Example: step functions

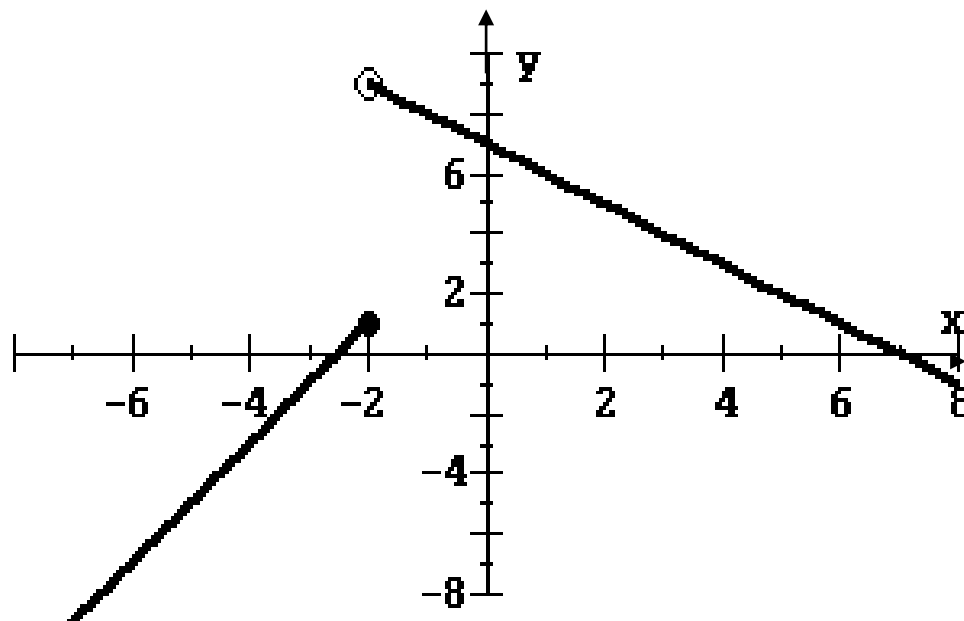
Consider the cost of travelling on the Rea Vaya bus. Here x is the number of km travelled and $C(x)$ is in rands:

$$C(x) = \begin{cases} 7.5 & \text{if } 0 \leq x \leq 5 \\ 9 & \text{if } 5 < x \leq 10 \\ 11 & \text{if } 10 < x \leq 15 \\ 13 & \text{if } 15 < x \leq 25 \\ 14 & \text{if } 25 < x \leq 35 \\ 15 & \text{if } 35 < x \end{cases}$$

Example

Sketch the graph of the function:

$$f(x) = \begin{cases} 2x + 5 & x \leq -2 \\ -x + 7 & x > -2 \end{cases}.$$



Examples

Sketch the following functions and find their domain and range:

(a) $f(x) = -3x + 4$

(b) $g(x) = x^2 - 2$

(c) $h(x) = \sec x$

Symmetry in functions

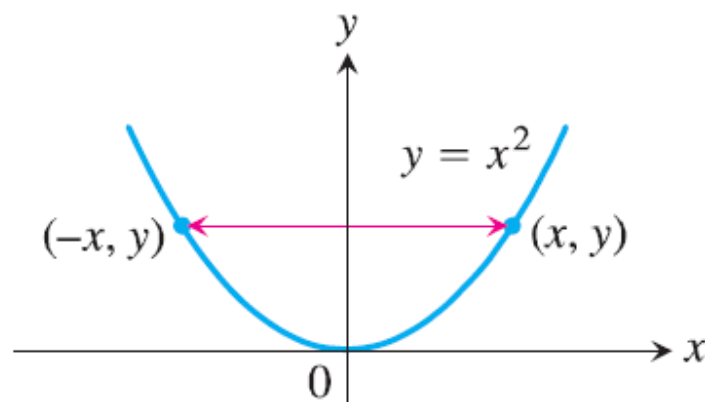
If a function f satisfies

$$f(-x) = f(x) \quad \text{for all } x \in D$$

then f is an **even** function.

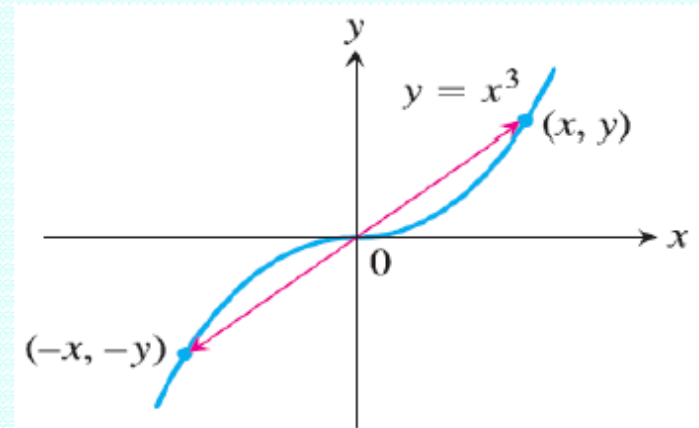
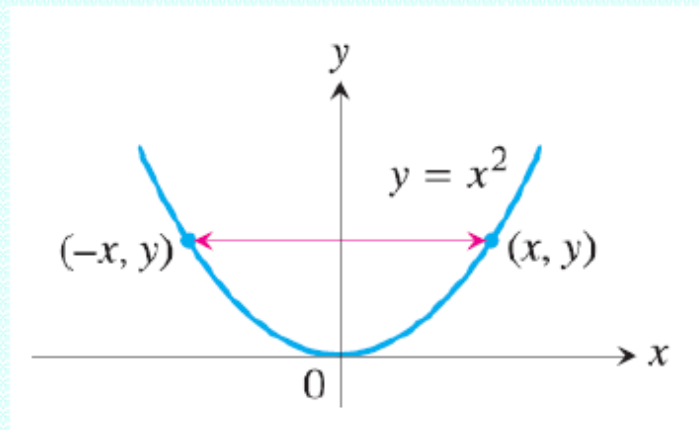
Examples:

- ▶ $f(x) = x^2$
- ▶ $f(x) = \cos(x)$
- ▶ $f(x) = |x|$



Another way of defining an even function is to say that it is a reflection about the y -axis.

- Sebuah fungsi f disebut **fungsi genap** bila $f(-x) = f(x)$.
- Grafik dari fungsi genap simetri terhadap sumbu y . Ω
- Sebuah fungsi f disebut **fungsi ganjil / gasal** bila $f(-x) = -f(x)$.
- Grafik dari fungsi gasal simetri terhadap titik $(0,0)$. Ω



Latihan:

- Periksa apakah fungsi berikut termasuk fungsi genap atau gasal.
 - $y = x^2$ Δ
 - $y = x^3$ Δ
 - $y = x^5 + 3x^2 + 1$ Δ
 - $y = |x|$
 - $y = |x - 1|$ Δ
 - $y = [x]$ Δ
 - $y = [x]$
 - $y = [x^2]$
- Adakah fungsi yang sekaligus fungsi genap dan gasal?

Increasing and decreasing functions

A function f is *increasing* on an interval I if whenever $x_1 < x_2$, we have $f(x_1) < f(x_2)$.

A function f is *decreasing* on an interval I if whenever $x_1 < x_2$ we have $f(x_1) > f(x_2)$.

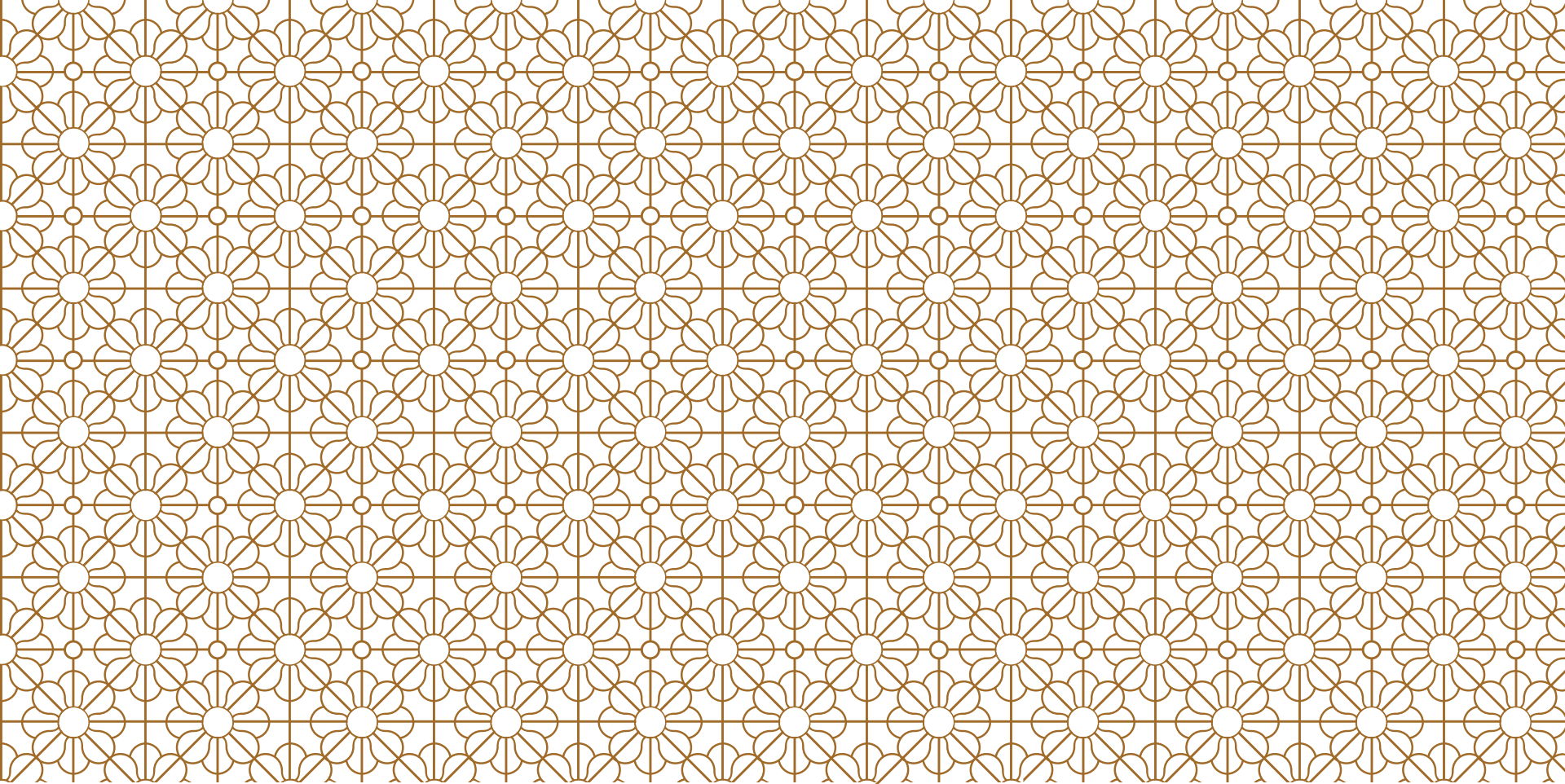
Example: Is the function $f(x) = \cos x$ increasing, decreasing, or neither over the following intervals:

(a) $x \in [\pi, 3\pi/2]$

(b) $x \in [0, \pi/2]$

(c) $x \in [\pi/2, 3\pi/2]$





FUNCTION

TYPES OF FUNCTIONS

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Polynomials

These are functions of the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

where $a_0, \dots, a_n$ are constants. If the leading coefficient $a_n \neq 0$ then $f(x)$ is a polynomial of degree n .

- ▶ polynomial of degree 1 = linear function
- ▶ polynomial of degree 2 = quadratic function
- ▶ polynomial of degree 3 = cubic function

Power functions

These are functions of the form

$$f(x) = x^a$$

Note that a can be

- ▶ a positive integer
- ▶ $a = \frac{1}{n}$ where n is a positive integer
- ▶ a negative integer, so $f(x) = \frac{1}{x^a}$

Rational functions

A rational function f is a ratio of two polynomials

$$f(x) = \frac{P(x)}{Q(x)}$$

Example:

$$f(x) = \frac{2x^4 - x^2 + 1}{x^2 - 4}$$

Exponential Functions

An **exponential function** is a function of the form

$$f(x) = a^x$$

where $a \in [0, \infty)$. If $x \in \mathbb{Z}^+$ (x is a positive integer), then $a^x = \underbrace{a \cdot a \cdot \dots \cdot a}_{x \text{ times}}$

If $x = 0$, then $a^x = 1$.

If $x \in \mathbb{Q}$, then $x = \frac{p}{q}$ where $p, q \in \mathbb{Z}, q \neq 0$. (If $x < 0$ we assume $q > 0$.) Then

$$a^x = a^{p/q} = \sqrt[q]{a^p} = (\sqrt[q]{a})^p$$

Consider $x \in \mathbb{R}$ such that x is rational. We can easily calculate a^x because $x = \frac{p}{q}$ for $p, q \in \mathbb{Z}, q \neq 0$. We get

$$a^x = a^{\frac{p}{q}} = \sqrt[q]{a^p}$$

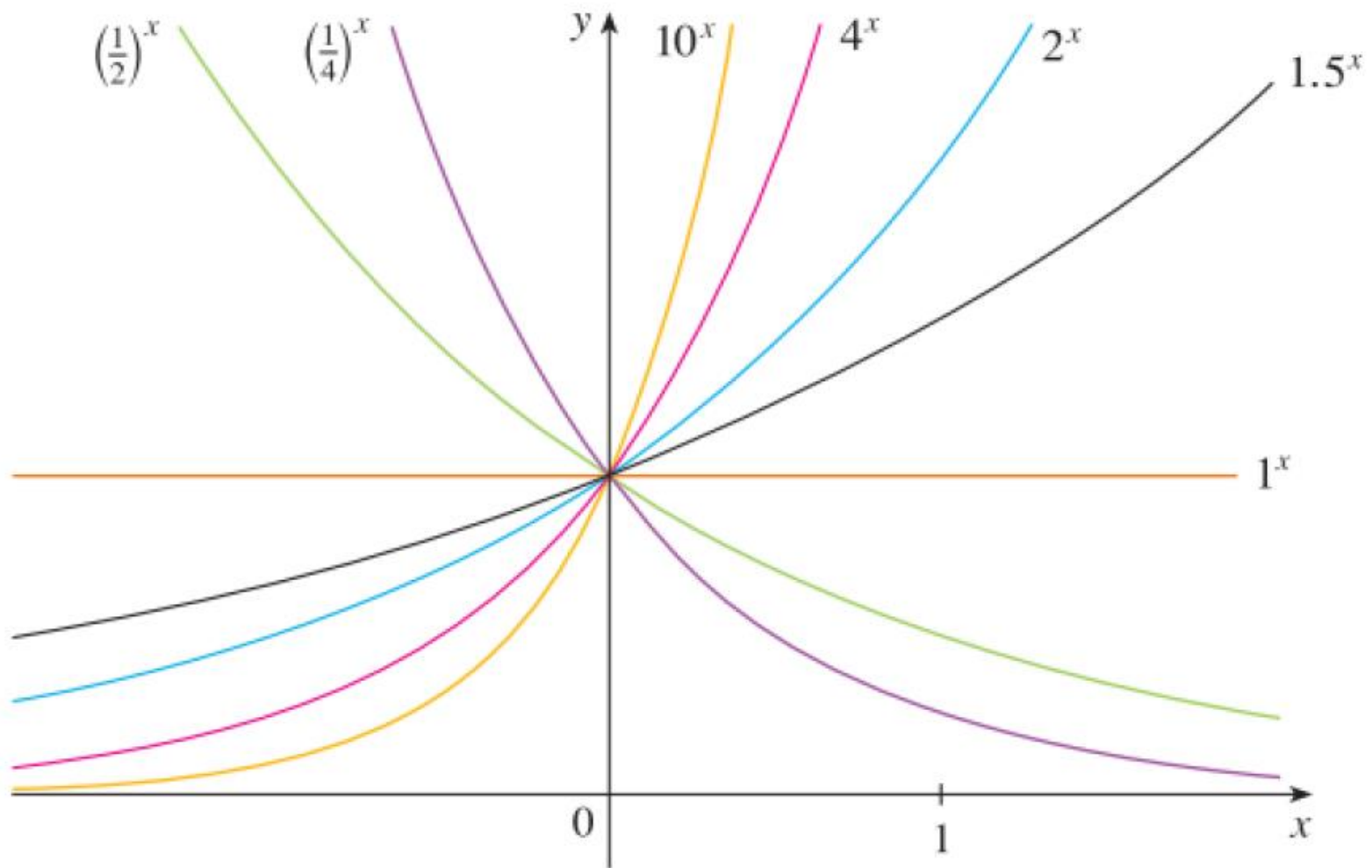
What about $2^{1.7}$ and $2^{1.8}$? We can write these as follows:

$$2^{1.7} = 2^{17/10} = \sqrt[10]{2^{17}}$$

and

$$2^{1.8} = 2^{18/10} = 2^{9/5} = \sqrt[5]{2^9}$$

Graphs of $y = a^x$



Exponent questions:

Rewrite and simplify the following expressions:

▶ $\frac{\sqrt{a}\sqrt{b}}{\sqrt[5]{ab}}$

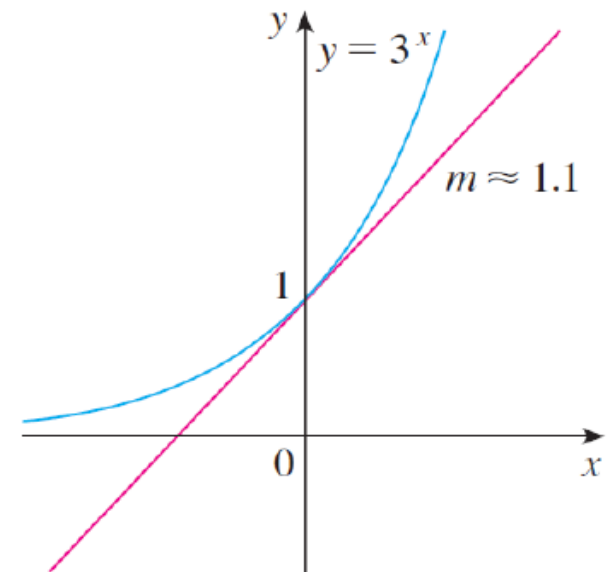
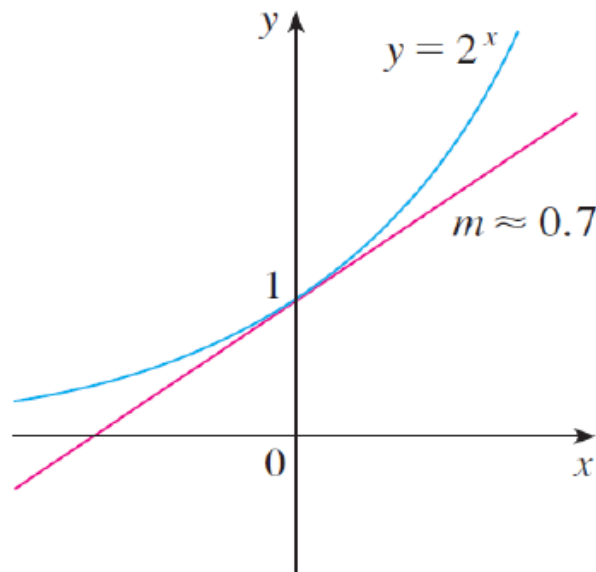
▶ $\frac{x^{4x} \cdot x^{5x+1}}{x^{x-5}}$

Exponential graph example:

Sketch $y = 4 - 5^x$.

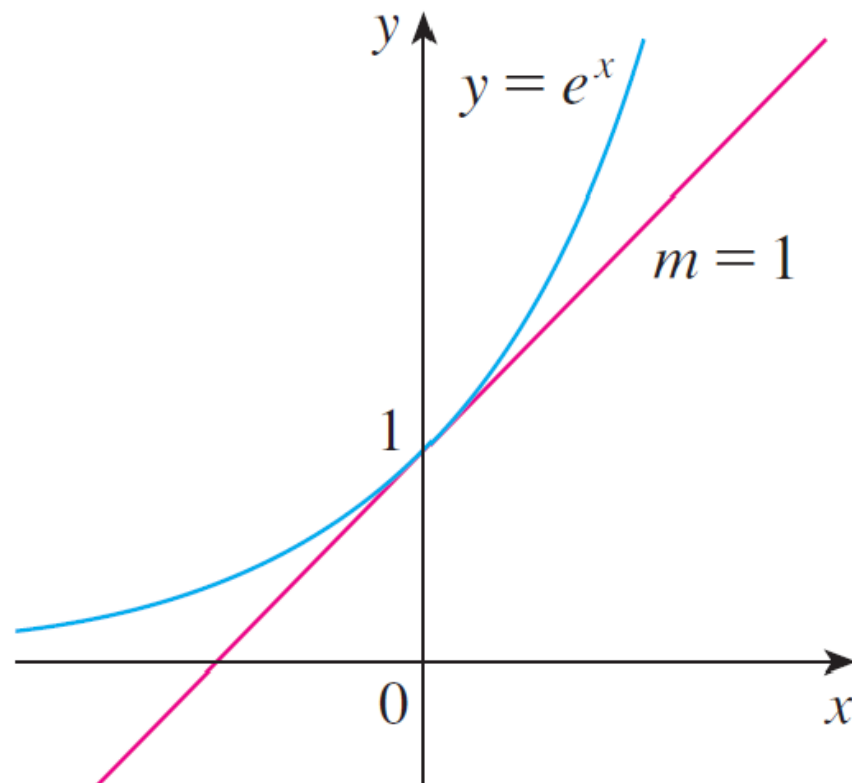
(Hint: write as $y = -5^x + 4$.)

The graph of $y = e^x$



The number e is the real number such that the gradient of the tangent to $y = e^x$ is equal to 1 when $x = 0$.

The graph of $y = e^x$



$$e \approx 2.71828$$

Laws of Exponents

If a and b are positive numbers and x and y are any real numbers, then

- ▶ $a^x \cdot a^y = a^{x+y}$
- ▶ $\frac{a^x}{a^y} = a^{x-y}$
- ▶ $(a^x)^y = a^{x \cdot y} = (a^y)^x$
- ▶ $(ab)^x = a^x \cdot b^x$

VERY IMPORTANT: for $a, b \neq 0$ and $x \neq 1$,

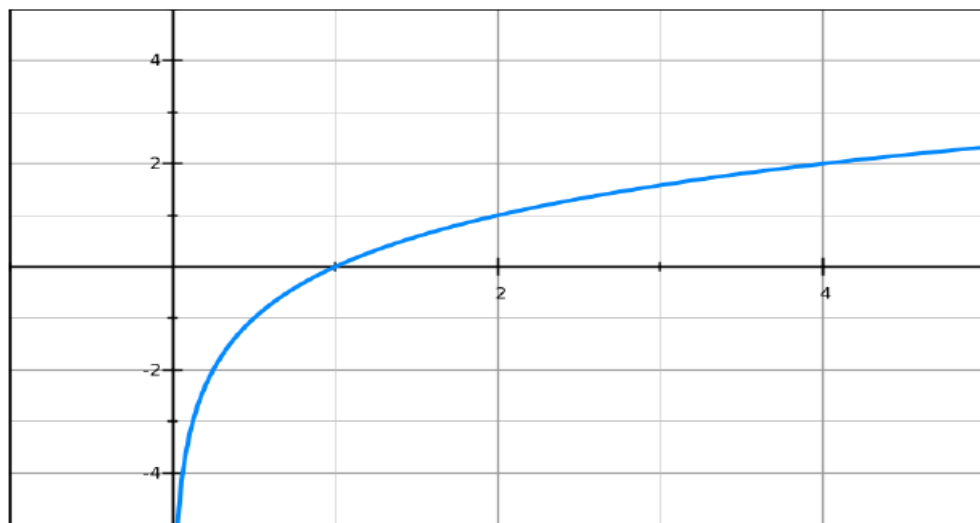
$$(a + b)^x \neq a^x + b^x$$

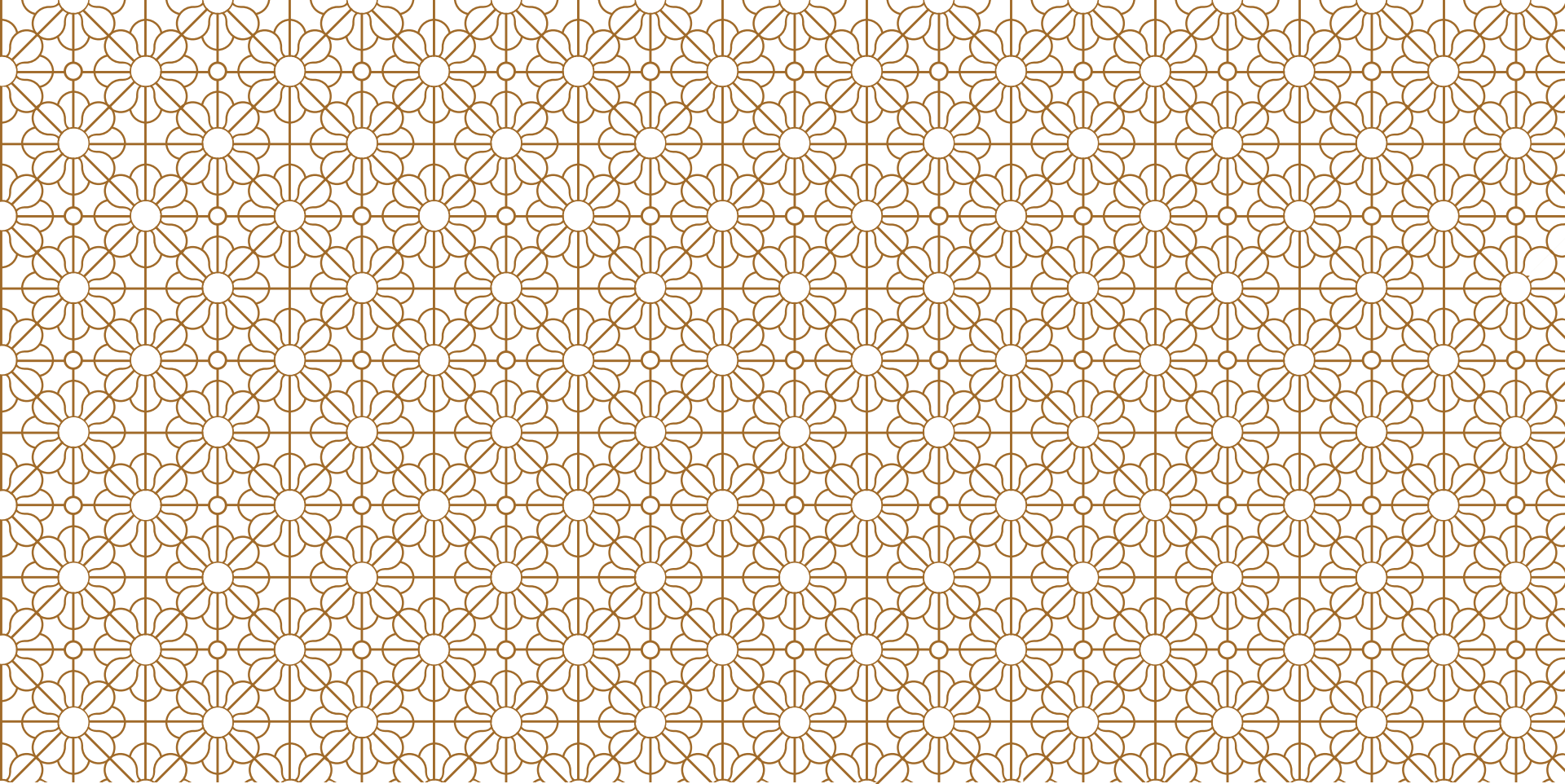
Logarithmic functions

$f(x) = \log_a x$ where $a \geq 0$ is a constant (also known as the *base*). Logarithmic functions are the inverse of exponential functions. That is,

$$\text{if } y = a^x \quad \text{then} \quad \log_a y = x.$$

$$f(x) = \log_2 x$$





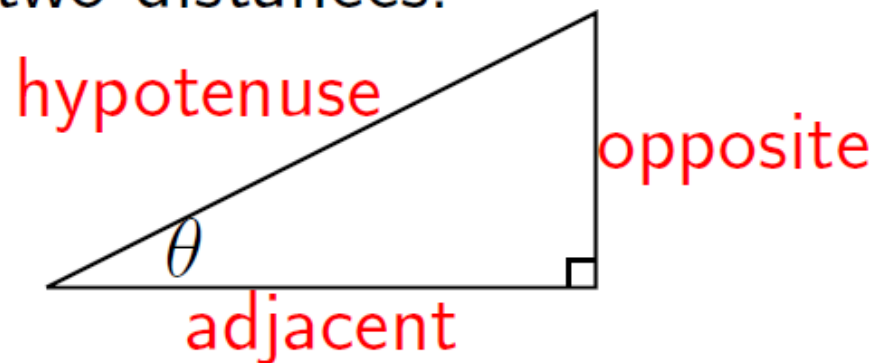
FUNCTION

TRIGONOMETRIC FUNCTIONS

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Trigonometric functions

Trig functions take as input an angle (measured in radians) and output the *ratio* between two distances.



$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

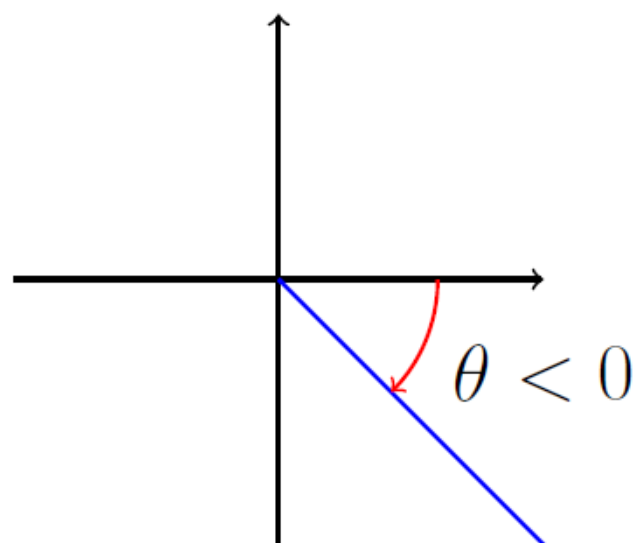
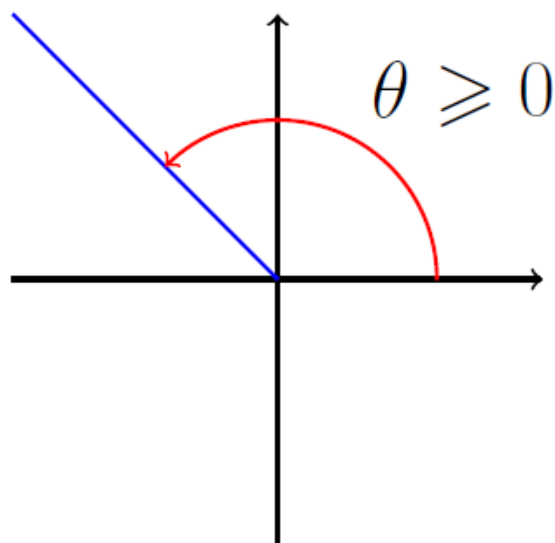
$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\csc \theta = \frac{\text{hyp}}{\text{opp}}$$

$$\sec \theta = \frac{\text{hyp}}{\text{adj}}$$

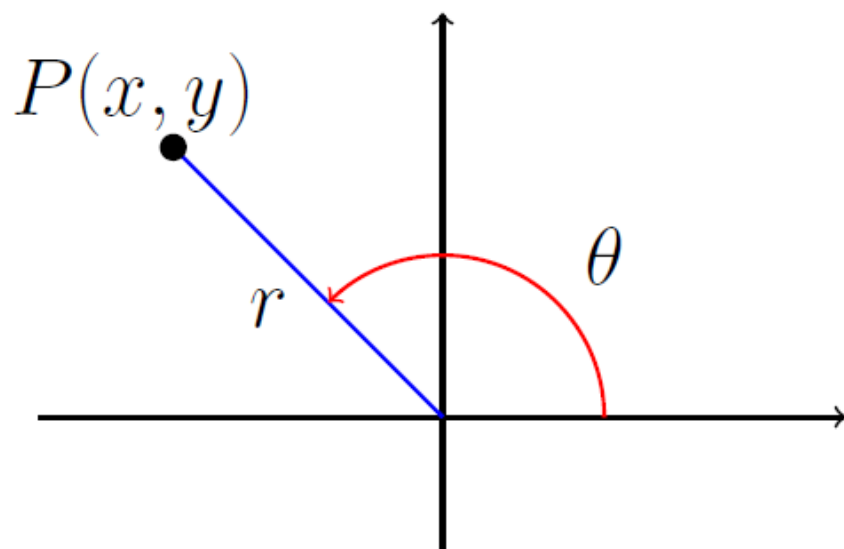
$$\cot \theta = \frac{\text{adj}}{\text{opp}}$$

More generally, angles can be measured in a coordinate system:



A positive and negative angle drawn in the standard position.

Trig functions in a coordinate system



$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

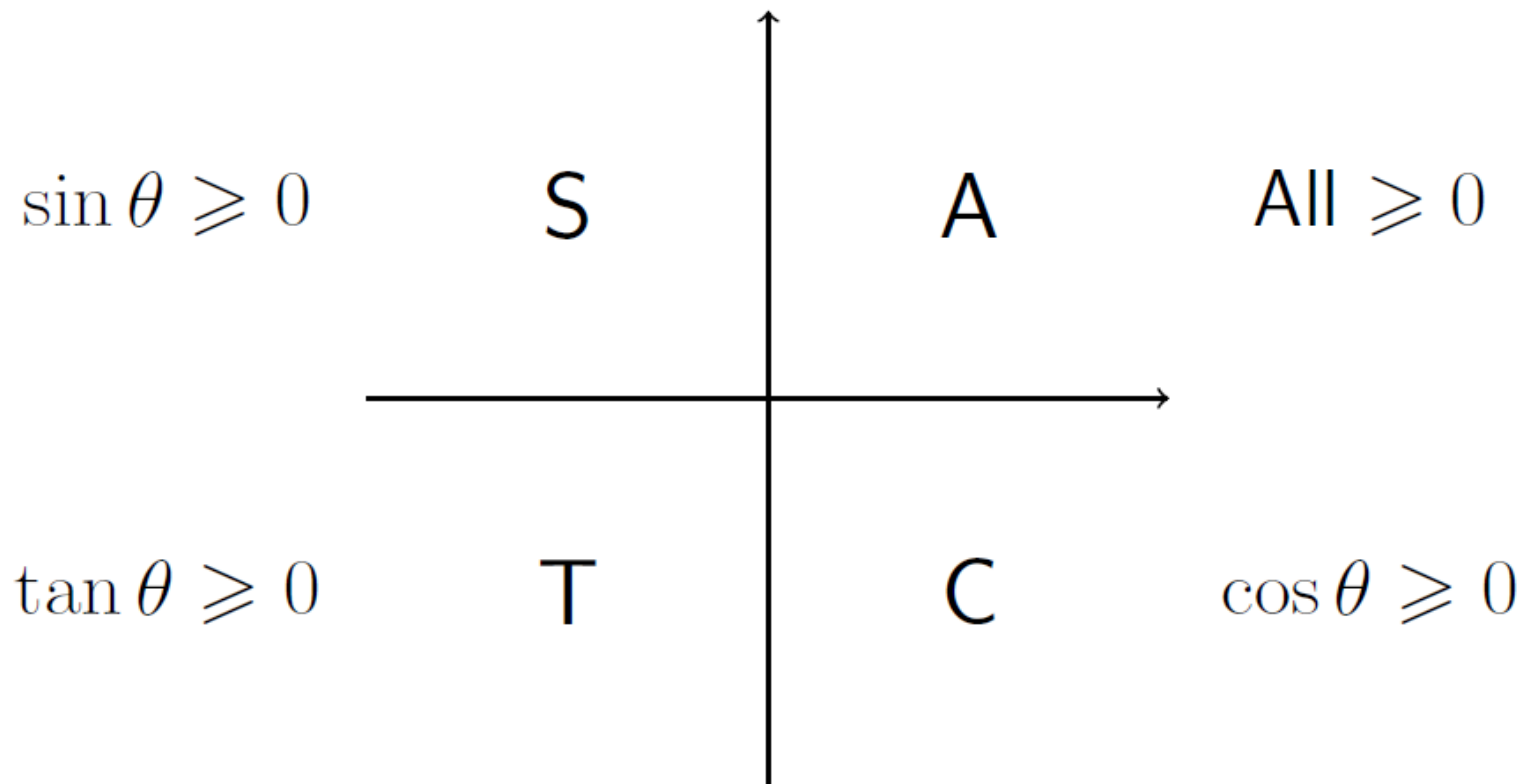
$$\tan \theta = \frac{y}{x}$$

$$\csc \theta = \frac{r}{y}$$

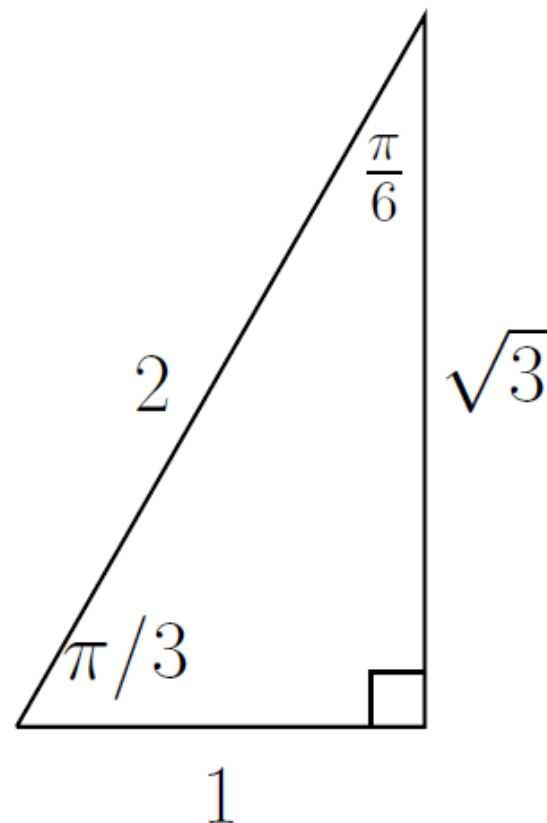
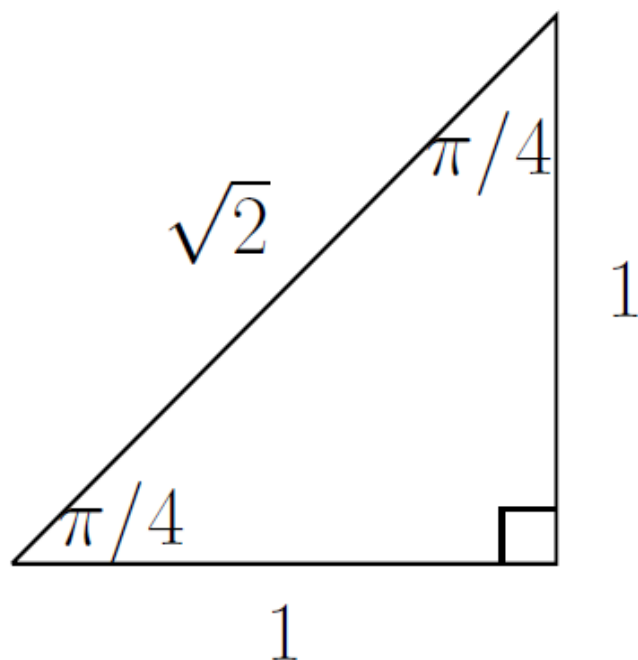
$$\sec \theta = \frac{r}{x}$$

$$\cot \theta = \frac{x}{y}$$

Signs of trig functions



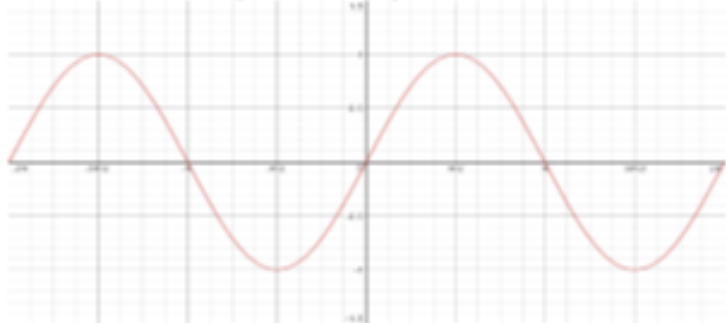
Special angles



Exercise: calculate all of the trig ratios for $\theta = 5\pi/3$.

Graphs of trig functions: $f(x) = \sin x$

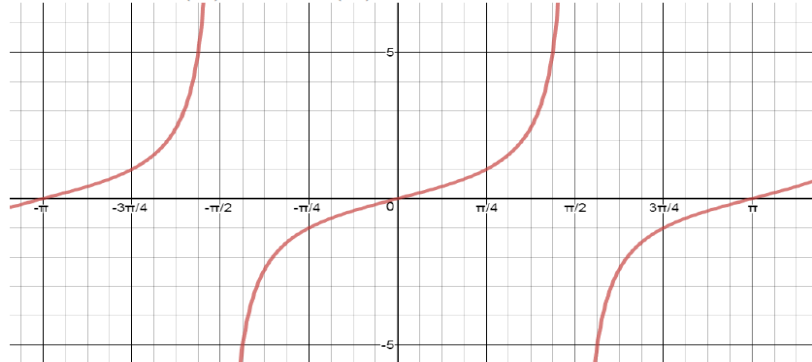
Note: $-1 \leq \sin x \leq 1$.



Graph of $f(x) = \cos(x)$. Again, $|\cos x| \leq 1$.

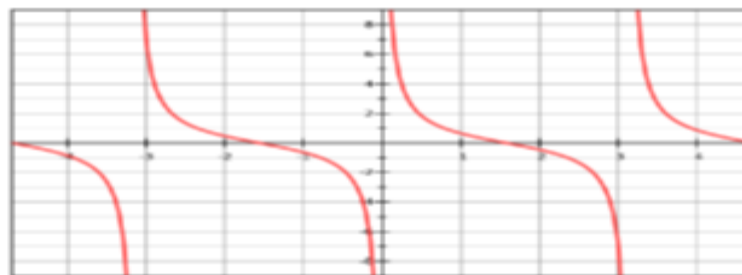


Graph of $f(x) = \tan(x)$.



Note the vertical asymptotes at $\pi/2$ and $-\pi/2$.

Graph of $f(x) = \cot(x)$.

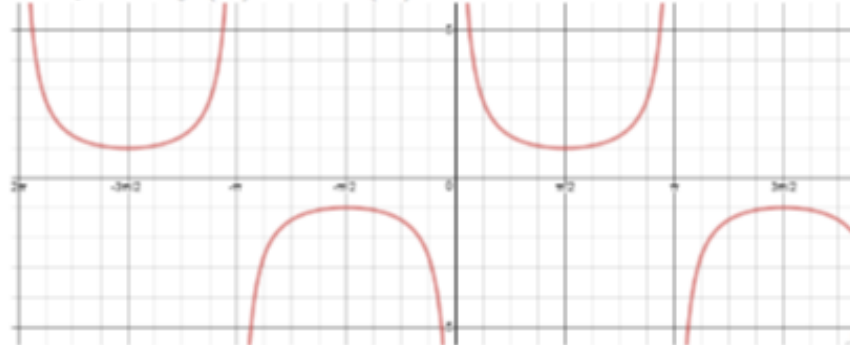


Note: $\cot(x) = 0$ wherever $\tan(x)$ has an asymptote.

Graph of $f(x) = \sec(x)$.



Graph of $f(x) = \csc(x)$.



Trig identities

Trig identities are useful relationships between trig functions. Some of the basic identities are:

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$$

More trig identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

Divide both sides by $\cos^2 \theta$ to get

$$\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$
$$\therefore \tan^2 \theta + 1 = \sec^2 \theta$$

Or, divide both sides of the original by $\sin^2 \theta$:

$$\frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$$
$$\therefore 1 + \cot^2 \theta = \csc^2 \theta$$

Addition formulas

$$\sin(x + y) = \sin x \cdot \cos y + \cos x \cdot \sin y$$

$$\cos(x + y) = \cos x \cdot \cos y - \sin x \cdot \sin y$$

How can we use

$$\sin(x + y) = \sin x \cdot \cos y + \cos x \cdot \sin y$$

to get a formula for

$$\sin(x - y)?$$

Result:

$$\sin(x - y) = \sin x \cdot \cos y - \cos x \cdot \sin y$$

We can also use the addition formulas to get the double-angle formulas:

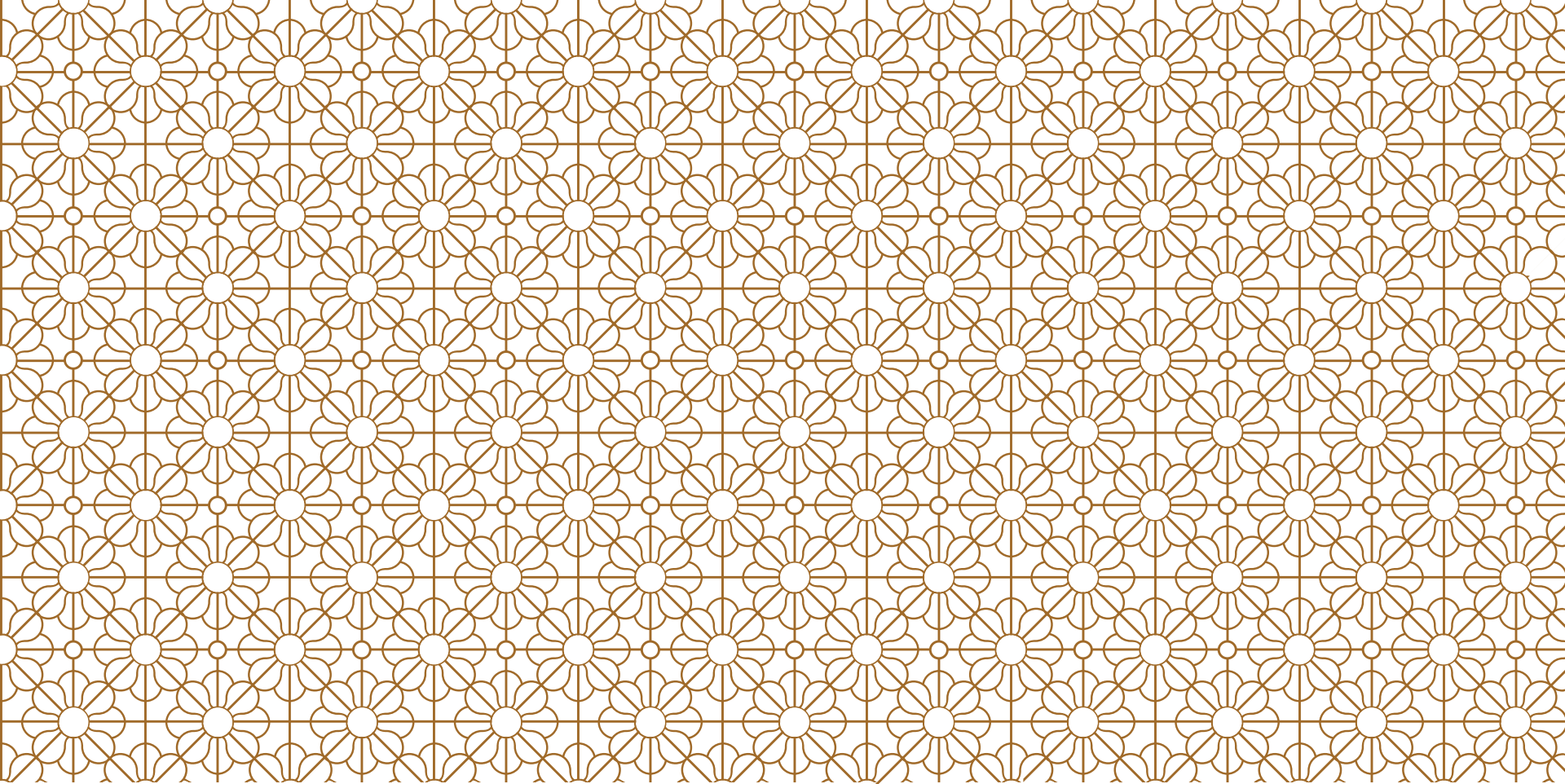
$$\sin(2x) = 2 \sin x \cdot \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$\cos(2x) = 2 \cos^2 x - 1$$

$$\cos(2x) = 1 - 2 \sin^2 x$$

Example: Find all of the values of x in the interval $[0, 2\pi]$ such that $\sin x = \sin 2x$.



FUNCTION

ALGEBRA OF FUNCTIONS

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Algebraic functions

An algebraic function is one that can be formed by using the algebraic operations of addition, subtraction, multiplication, powers, division and taking roots.

Note: any rational function is automatically an algebraic function.

Examples:

$$f(x) = \sqrt{x^2 + 1} \qquad h(x) = \frac{x^4 - 16x^2}{x + \sqrt{x}}$$

Misalkan f dan g dua buah fungsi real dengan daerah definisi D_f dan D_g

▪ $(f + g)(x) = f(x) + g(x)$ Ω

$$D_{f+g} = D_f \cap D_g$$

▪ $(f - g)(x) = f(x) - g(x)$ Ω

$$D_{f-g} = D_f \cap D_g$$

▪ $(fg)(x) = f(x) \cdot g(x)$ Ω

$$D_{fg} = D_f \cap D_g$$

▪ $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ Ω

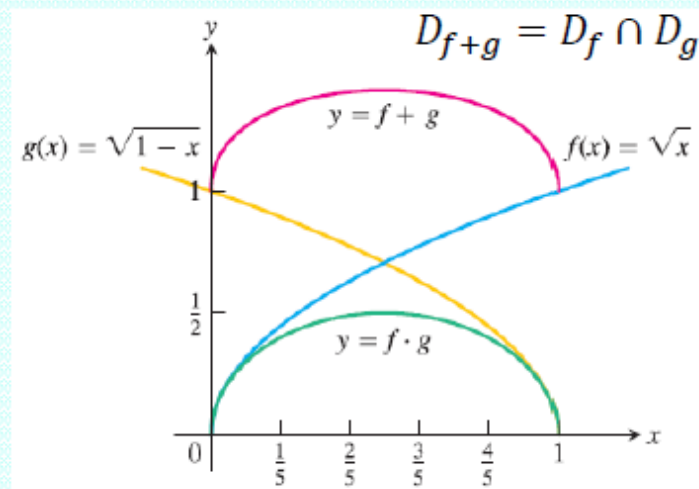
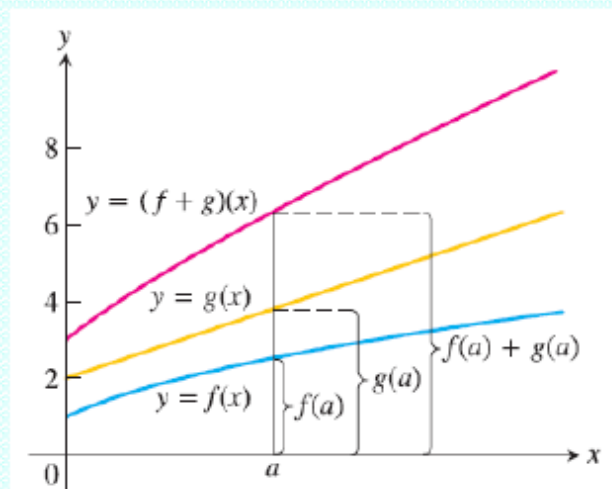
$$D_{\frac{f}{g}} = D_f \cap D_g \cap \{x \mid g(x) \neq 0\}$$

▪ $(cf)(x) = c \cdot f(x)$ Ω

$$D_{cf} = D_f \quad (c \text{ konstanta})$$

▪ $f^n(x) = \underbrace{f(x) \cdot f(x) \cdots f(x)}_{n \text{ suku}}$

$$D_{f^n} = D_f \quad (n \text{ bilangan asli})$$



Domain of a Function

Example 1: Find the domain of $f(x) = \sqrt{7x+5}$

Since the square root of a negative number is undefined we must have

$$7x+5 \geq 0 \Rightarrow 7x \geq -5 \Rightarrow x \geq -\frac{5}{7}.$$

The domain is all values of x that satisfy $x \geq -\frac{5}{7}$.

The domain can be expressed as the interval $\left(-\frac{5}{7}, \infty\right)$

Example 2: Find the domain of $f(x) = \frac{2x}{3x-1}$

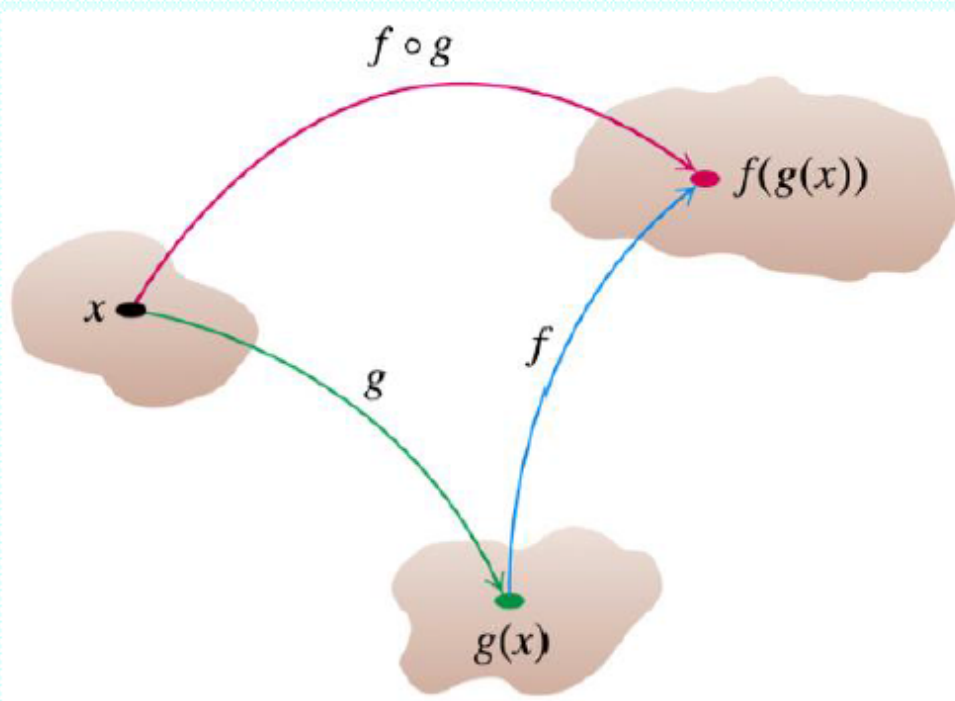
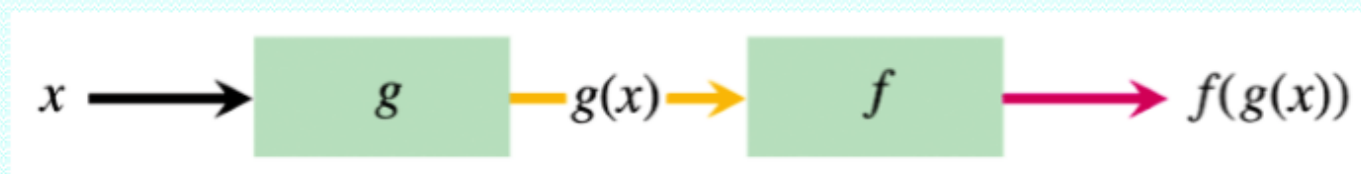
Since division by zero is undefined we must

$$\text{have } 3x - 1 \neq 0 \Rightarrow x \neq \frac{1}{3}.$$

The domain can be expressed as the intervals

$$\left(-\infty, \frac{1}{3}\right) \text{ and } \left(\frac{1}{3}, \infty\right)$$

- Misalkan f dan g dua buah fungsi. Fungsi komposisi f dengan g , dinotasikan $f \circ g$ didefinisikan sebagai $(f \circ g)(x) = f(g(x))$



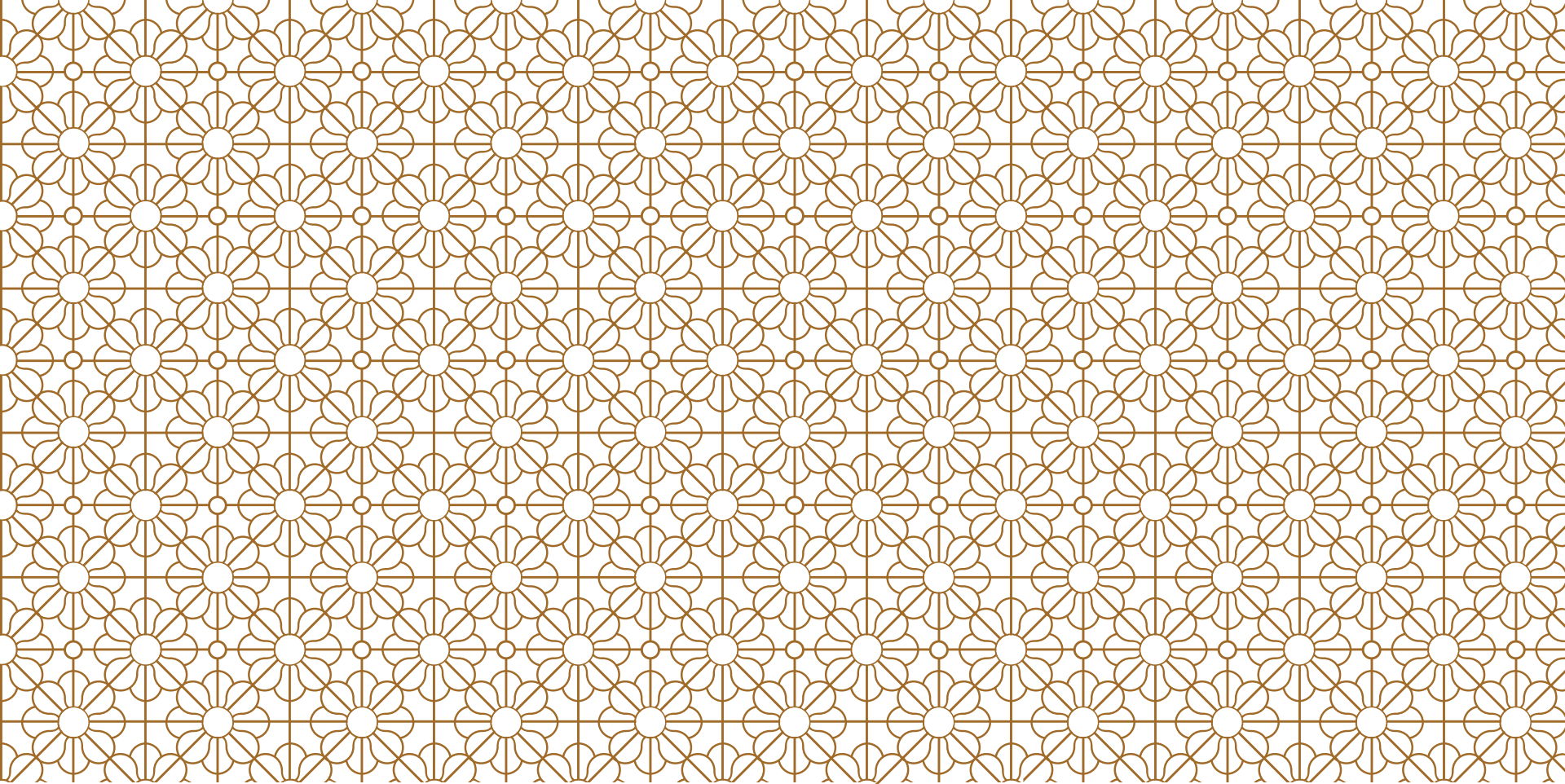
Daerah definisi fungsi komposisi $f \circ g$ adalah semua titik pada D_g yang bersifat $g(x) \in D_f$

Contoh:

$$\begin{aligned}
 f(x) &= x^2 \text{ dan } g(x) = \sqrt{x-1}. \\
 (f \circ g)(x) &= f(g(x)) = (g(x))^2 \\
 &= (\sqrt{x-1})^2 = |x-1|
 \end{aligned}$$

$$D_{f \circ g} = [1, \infty)$$

Mengapa $D_{f \circ g} \neq \mathbb{R}$?



FUNCTION

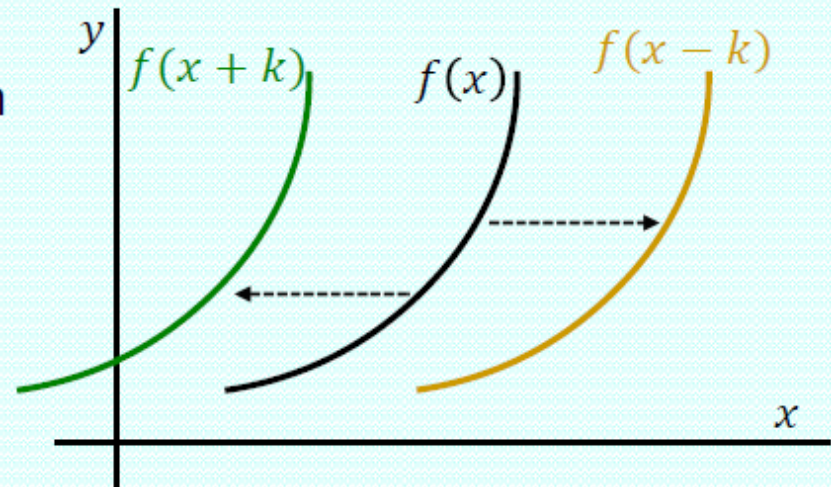
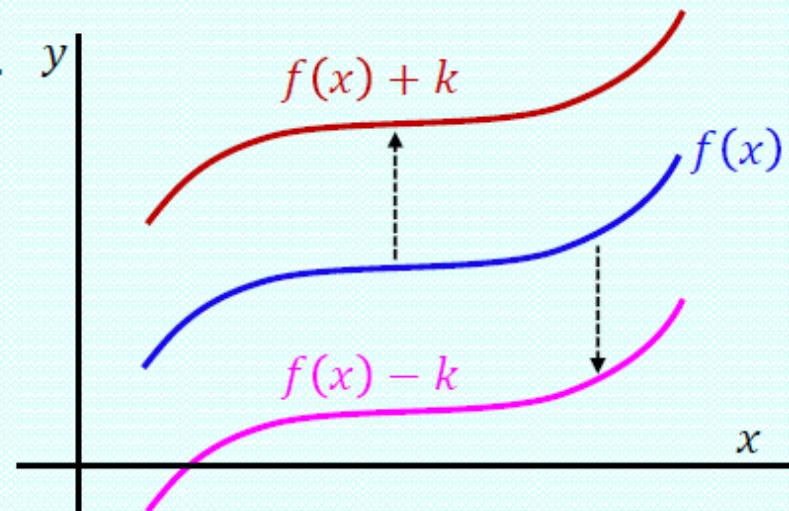
TRANSFORMATION OF FUNCTIONS

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Pergeseran Grafik Fungsi

Diberikan grafik fungsi real $y = f(x)$ dan $k > 0$.

- Grafik $y = f(x) - k$ merupakan pergeseran dari grafik $f(x)$ sejauh sejauh k ke bawah.
- Grafik $y = f(x) + k$ merupakan pergeseran dari grafik $f(x)$ sejauh sejauh k ke atas.
- Grafik $y = f(x - k)$ merupakan pergeseran dari grafik $f(x)$ sejauh k ke kanan. Δ
- Grafik $y = f(x + k)$ merupakan pergeseran dari grafik $f(x)$ sejauh k ke kiri.
- Latihan:
Gambarlah grafik $g(x) = x^2 + 4x + 3$ berdasarkan pergeseran grafik $f(x) = x^2$



Peregangan, Pemampatan, dan Pencerminkan Grafik Fungsi

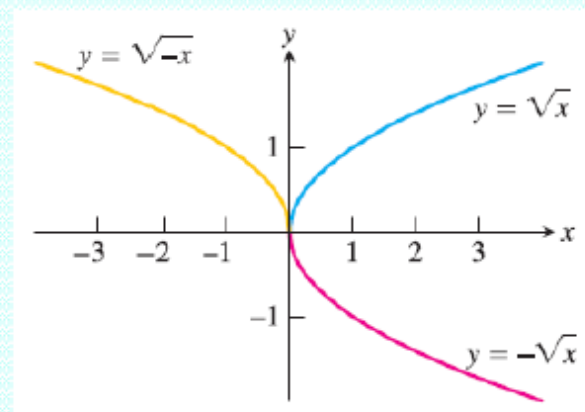
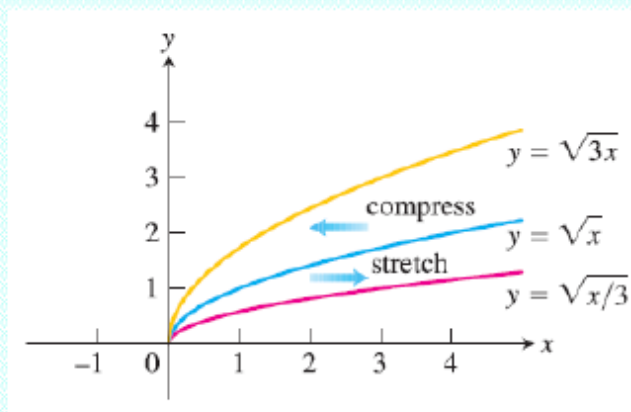
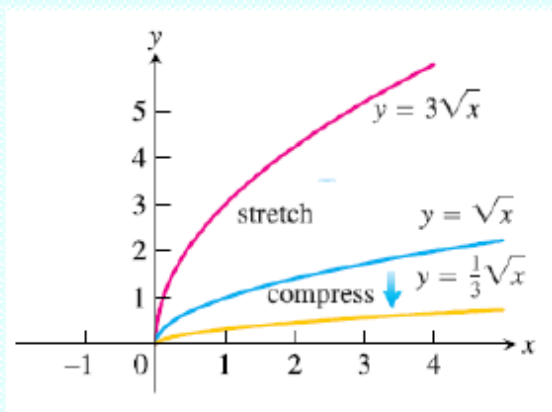
Diberikan grafik fungsi real $y = f(x)$

■ Untuk $c > 1$, berlaku sifat-sifat berikut:

- $y = cf(x)$ Peregangan grafik f pada arah vertikal dengan faktor c .
- $y = \frac{1}{c}f(x)$ Pemampatan grafik f pada arah vertikal dengan faktor c .
- $y = f(cx)$ Pemampatan grafik f pada arah horizontal dengan faktor c .
- $y = f\left(\frac{x}{c}\right)$ Peregangan grafik f pada arah horizontal dengan faktor c .

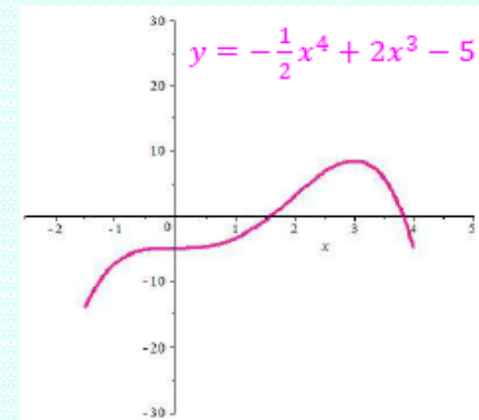
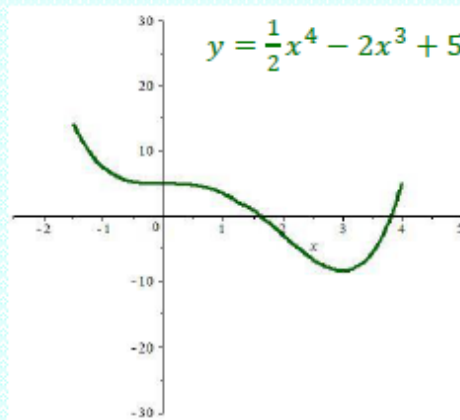
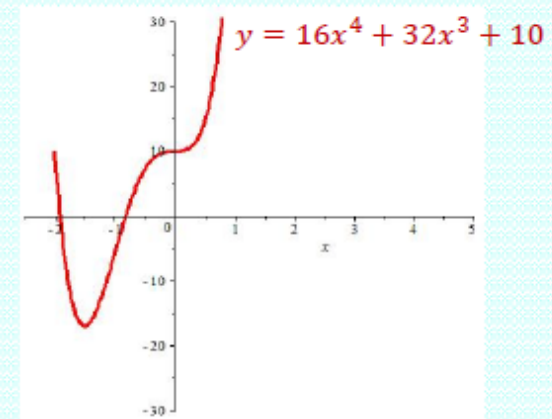
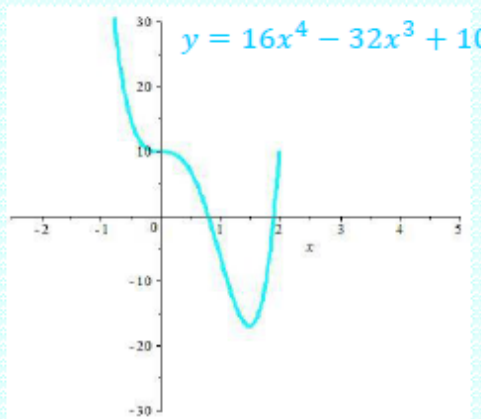
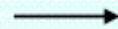
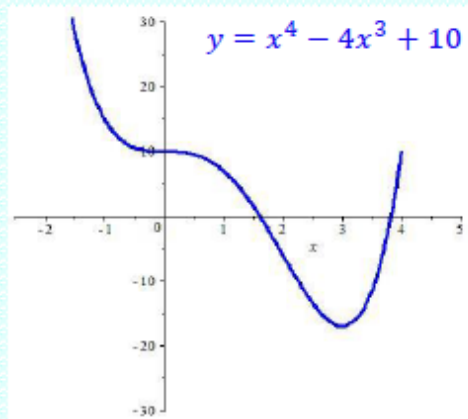
■ Untuk $c = -1$, berlaku sifat-sifat berikut:

- $y = -f(x)$ Pencerminkan grafik f terhadap sumbu x .
- $y = f(-x)$ Pencerminkan grafik f terhadap sumbu y .



Misalkan $f(x) = x^4 - 4x^3 + 10$. Tentukan formula dan gambarkan grafiknya bila:

- Fungsi f dimampatkan secara horizontal dengan faktor 2, kemudian dicerminkan terhadap sumbu y .
- Fungsi f dimampatkan secara vertical dengan faktor 2, kemudian dicerminkan terhadap sumbu x .



Examples:

Sketch the graph of each of the following functions:

- ▶ $g(x) = \cos(2x), x \in [0, 2\pi]$

- ▶ $h(x) = 2 - \cos(x), x \in [-\pi, \pi]$

Hint: re-write as $h(x) = -\cos(x) + 2$

- ▶ $f(x) = x^2 - 4x + 7$

Hint: complete the square

- ▶ $v(\theta) = \tan(\theta + \pi/2), \theta \in [-\pi, \pi]$

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